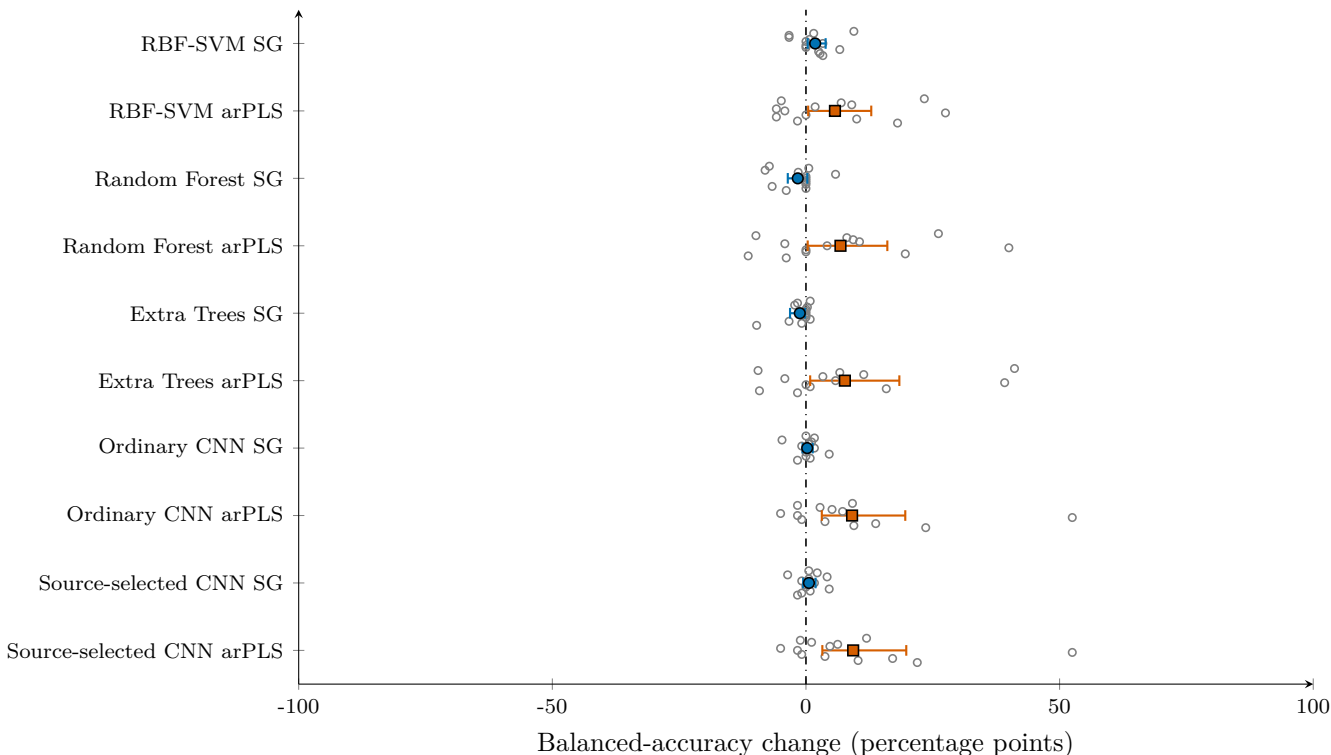


S: M06 combined predictions per physical sample | instrument-only



RQ-S01 / P08-F03 universal preprocessing effect

Aggregation: primary: equally weighted contexts within domain then equal domains. Interval weighting: instrument-only. Positive values mean the universal preprocessing pipeline helps relative to the MIN baseline. MIN uses minimal min-max scaling; SG combines impulse replacement and Savitzky-Golay smoothing; arPLS combines impulse replacement and baseline correction. All three finish with min-max scaling to $[0,1]$. The compared policies differ by the whole preprocessing pipeline, not by an isolated smoothing or background operation. Gray points are the 13 station/instrument domains (not samples); colored points are the equal-domain effect with 95% marginal conditional percentile intervals (10,000 support-preserving weights). M01 uses individual predictions; M06 averages MODEL probabilities within sample/instrument then equal instruments, not mean spectra. No retraining, no new instrument uncertainty, no G4 interval, no superiority claim; source-only model selection/calibration, frozen CNN identity across policies. This figure is not clean-spectrum recovery. 598 original spectra, 69 physical samples, 10 instruments, 557 held spectra, 260 contexts.